Assignment - 3

PHY 111

Sec - 22

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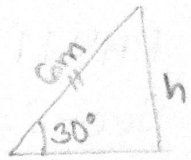
1. (a) From, Fig-1 we can say,

$\sin 30^\circ = \frac{h}{6}$



$\Rightarrow \frac{1}{2} = \frac{h}{6}$

$\Rightarrow h = 3m$

mass, $m = 20kg$

$u = 0ms^{-1}$

$v = ?$

$\therefore v^2 = u^2 + 2gh$

$\Rightarrow v^2 = 0^2 + 2 \times 9.8 \times 3$

$\Rightarrow v = \sqrt{2 \times 9.8 \times 3}\ ms^{-1}$

$\Rightarrow v = 7.67\ ms^{-1}$

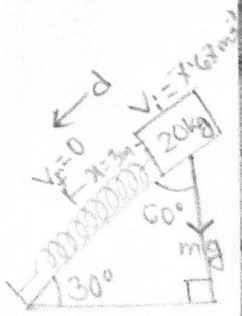
$\therefore$ The speed of the block just before it hits the spring is $v = 7.67\ ms^{-1}$.

(b) We know,

$W_{net} = \Delta K.E.$

$\Rightarrow mgd\cos 60^\circ - \frac{1}{2}Kx^2 = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$\Rightarrow 20 \times 9.8 \times 3 \times \frac{1}{2} - \frac{1}{2}K(3)^2 = \frac{1}{2} \times 20 \times 0^2 - \frac{1}{2} \times 20 \times (7.67)^2$



$\Rightarrow 294 - \frac{9K}{2} = -588$

$\Rightarrow -\frac{9K}{2} = -588 - 294$

$\Rightarrow -\frac{9K}{2} = -882$

$\Rightarrow -9K = -1764$

$\Rightarrow K = \frac{1764}{9}\ Nm^{-1}$

$\Rightarrow K = 196\ Nm^{-1}$

$m = 20kg$

$g = 9.8ms^{-2}$

$d = 3m$

$x = 3m$

$v_f = 0ms^{-1}$

$v = v_i = 7.67ms^{-1}$

$K = ?$

$\therefore$ The spring constant, $K = 196\ Nm^{-1}$

2. (a) From, Fig-2 we get,

$W_{net} = \Delta K.E.$

$\Rightarrow -\frac{1}{2}Kx^2 = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_c^2$

$\Rightarrow -\frac{1}{2}Kx^2 = 0 - \frac{1}{2}mv_c^2$

$\Rightarrow v_c^2 = \frac{Kx^2}{m}$

$\Rightarrow v_c = \sqrt{\frac{Kx^2}{m}}$

$\Rightarrow v_c = \sqrt{\frac{2250 \times (0.3)^2}{10}}\ ms^{-1}$

$\Rightarrow v_c = 4.5\ ms^{-1}$

$m = 10kg$
$v_f = 0ms^{-1}$
$v_c = ?$
$K = 2250\ Nm^{-1}$
$x = 0.300m$
$= 0.3m$

$\therefore$ The speed of the block at point c is $4.5\ ms^{-1}$.

(b) $v_B^2 = u^2 + 2gh$

$\Rightarrow v_B = \sqrt{u^2 + 2gh}$

$\Rightarrow v_B = \sqrt{0^2 + 2 \times 9.8 \times 3}\ ms^{-1}$

$\Rightarrow v_B = 7.67\ ms^{-1}$

$u = 0ms^{-1}$
$g = 9.8\ ms^{-2}$
$h = 3m$
$f_K = \mu_K R$
$= \mu_K mg$
$d = BC = 6m$
$m = 10kg$
$v_c = 4.5\ ms^{-1}$
$\mu_K = ?$

$\therefore W_{BC} = \Delta K.E.$

$\Rightarrow f_K d \cos 180^\circ = \frac{1}{2}mv_c^2 - \frac{1}{2}mv_B^2$

$\Rightarrow -\mu_K mgd = \frac{1}{2}m(v_c^2 - v_B^2)$

$\Rightarrow -\mu_K gd = \frac{1}{2}\{(4.5)^2 - (7.67)^2\}$

$\Rightarrow -\mu_K g d = -19.28$

$\Rightarrow \mu_K = \dfrac{19.28}{9.8 \times 6}$

$\Rightarrow \mu_K = 0.328$

$\therefore$ The coefficient of kinetic friction between the block and the rough surface between B and C is 0.328.

3.(a) We know that,

$m_1 = 70\,kg$
$u_1 = 0\,ms^{-1}$
$m_2 = 35\,kg$
$u_2 = 0\,ms^{-1}$
$v_1 = 0.55\,ms^{-1}$
$v_2 = ?$

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$\Rightarrow 70 \times 0 + 35 \times 0 = 70 \times 0.55 + 35 \times (-v_2)$

$\Rightarrow 0 = 38.5 - 35v_2$

$\Rightarrow 35v_2 = 38.5$

$\Rightarrow v_2 = \dfrac{38.5}{35}$

$\Rightarrow v_2 = 1.1\,ms^{-1}$

$\therefore$ The initial speed of the son after the push is $1.1\,ms^{-1}$.

We know that,

(b) $J = \Delta P$

$\Rightarrow J = P_f - P_i$

$\Rightarrow J = m_2 v_f - m_2 v_i$

$\Rightarrow J = 35 \times (-1.1) - 35 \times 0$

$\Rightarrow J = -38.5$

$m_2 = 35\ kg$

$u_2 = v_i = 0\ ms^{-1}$

$v_2 = v_f = -1.1\ ms^{-1}$

$t = 0.6\ s$

$F_{avg} = ?$

Now, $J = F_{avg} t$

$\Rightarrow F_{avg} = \frac{J}{t}$

$\Rightarrow F_{avg} = \frac{-38.5}{0.6}$

$\Rightarrow F_{avg} = -64.17\ N$

The negative sign indicates that the force is exerted in the opposite direction to the son's initial motion. So, the magnitude of the average force exerted on the son by the mother during the push is 64.17 N.